



Air pollution and noise emission monitoring in the port areas of the Balearic Islands

CLIENT

Balearic Port Authority (APB)

SECTOR

Ports

LOCATION

Balearic Islands, Spain

PRODUCTS

Kunak AIR Pro
Kunak AIR Cloud

Need

The APB, in its goal to intensify the concern for the environment, has designed an environmental management system certified by AENOR according to ISO 14001: 2015.

Its objective was "to monitor the air quality by measuring the pollutant emissions in the air of the port area as well as the noise levels".

This monitoring will allow them to know the origins and magnitudes of the pollutant and noise sources to study proposals that contribute to improving the quality of life of users and citizens.

With the aim of "monitoring air quality by measuring pollutant emissions in the air of the port area and acoustic levels", the Balearic Islands Port Authority sought to design a network of air quality stations

in the ports of Palma, Eivissa, Maó, Alcudia and La Savina.

This monitoring network will allow them to know the origins and magnitudes of pollutant and noise sources in order to study proposals that contribute to improving air quality and the quality of life of users and citizens.





The APB wanted to improve the operation and sustainability of its operations with an air quality network in the ports of the Balearic Islands.



Solution

Deployment of a network of 25 Kunak AIR Pro stations: 8 in the port of Palma, 6 in Ibiza, 4 in Maó, 4 in Alcudia and 3 in La Savina for monitoring CO, O₃, NO, NO₂ y SO₂ and PM₁, PM_{2.5}, PM₄, PM₁₀, Total Suspended Particles (TSP) and Total Particle Count (TPC) and noise pollution in real time in 8 ports of the Balearic Islands.

The solar panel-powered network also collects data on environmental parameters such as temperature, relative humidity and atmospheric pressure, rainfall and wind speed and direction.

The data is collected and sent to a cloud platform available to the APB.

Results

- SIAM environmental data management platform for the execution of pollutant dispersion models and rollback analysis.
- Data analysis for the implementation of improvement plans.
- Real-time air quality information panel integrated into the port authority's website.
- Kunak AIR Cloud platform for operational management of the sensor network.

Benefits

Development of a calibrated environmental monitoring solution for ports with the following functionalities:

- Analysis of the incidence of pollutants originating in the port.
- Alarm, alerts and statistics system in real time.
- Design of predictive and back-trajectory models for the evaluation of pollution dispersion.
- Periodic report generation system.
- Detection of episodes of increased pollution.
- Remote management and calibration of equipment.
- Development of advanced statistical protocols for obtaining high quality and reliable data that provide high correlations with official stations and reference equipment.

